

Riemann Integration

Comprehensive Cheat Sheet & Key Results

Based on "Riemann Integration.pdf" Chapter 4 Notes

1. Core Definitions

Partition: Let f be defined on closed interval $[a, b]$. A partition P is a finite ordered set of points:

$$P = \{x_0, x_1, \dots, x_n : a = x_0 < x_1 < \dots < x_n = b\}$$

Norm of a Partition ($\|P\|$): For each $i = 1, 2, \dots, n$, let $\Delta x_i = x_i - x_{i-1}$ be the length of the subinterval $[x_{i-1}, x_i]$. The norm or mesh is:

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i$$

Refinement: A partition P^* is called a refinement of P if $P \subset P^*$.

Common Refinement: If P_1 and P_2 are partitions of $[a, b]$, then $P = P_1 \cup P_2$ is a common refinement of P_1 and P_2 .

2. Riemann Sums

Let f be a bounded real-valued function on $[a, b]$ and P be a partition. Let:

$$M_i = \sup_{x_{i-1} \leq x \leq x_i} f(x), \quad m_i = \inf_{x_{i-1} \leq x \leq x_i} f(x)$$

Upper & Lower Sums

Upper Riemann Sum:

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i$$

Lower Riemann Sum:

$$L(P, f) = \sum_{i=1}^n m_i \Delta x_i$$

Oscillatory Sum:

$$W(P, f) = U(P, f) - L(P, f)$$

3. The Riemann Integral

Upper Riemann Integral:

$$\int_a^b f(x) dx = \inf_P U(P, f)$$

Lower Riemann Integral:

$$\int_a^b f(x) dx = \sup_P L(P, f)$$

Riemann Integrability Condition

A bounded real-valued function f on $[a, b]$ is said to be Riemann integrable if the upper and lower Riemann integrals are equal. The common value is denoted by $\int_a^b f(x) dx$.

$$\int_a^b f(x) dx = \overline{\int_a^b f(x) dx} = \underline{\int_a^b f(x) dx}$$

If f is Riemann integrable on $[a, b]$, we write $f \in R[a, b]$.

4. Key Results & Theorems

If f is a bounded function on $[a, b]$ and P is a partition, then:

$$m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a)$$

Furthermore:

$$m(b-a) \leq \int_a^b f(x) dx \leq \overline{\int_a^b f(x) dx} \leq M(b-a)$$

Functions that are Riemann Integrable ($f \in R[a, b]$):

1. Every **constant function** on $[a, b]$.
2. Every **continuous function** on $[a, b]$.
3. Every **monotonic function** on $[a, b]$.
4. A bounded function with only **finitely many** points of discontinuity.
5. A bounded function where the set of discontinuity points is **countable**.
6. A bounded function where the set of discontinuity points has a **finite number of limit points**.

Partition Inequality: If P_1, P_2 are partitions of $[a, b]$, then $L(P_1, f) \leq U(P_2, f)$, which deduces $\underline{\int} \leq \overline{\int}$.

5. Integral as a Limit of a Sum

Definition: Let $S(P, f) = \sum_{i=1}^n f(t_i) \Delta x_i$, where $t_i \in [x_{i-1}, x_i]$, be a Riemann sum. Then:

$$\lim_{\|P\| \rightarrow 0} S(P, f) = I$$

if $\forall \epsilon > 0, \exists \delta > 0$ such that $|S(P, f) - I| < \epsilon$ for every Riemann sum with $\|P\| < \delta$.

Definite Integral as Series Limit

If $f \in R[a, b]$, then:

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n hf(a + ih)$$

where $nh = b - a$.

Linearity Properties:

- If $f, g \in R[a, b]$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g \in R[a, b]$.
- $f - g \in R[a, b]$ and $\int (f - g)dx = \int f dx - \int g dx$.

6. Solved Examples

Example 1 (Countable Discontinuities):

Let f on $[0, 1]$ be defined by $f(x) = 1$ for $x \neq 1/2$ and $f(x) = 0$ for $x = 1/2$.

Solution: Discontinuous at only one point (countable). Thus, $f \in R[0, 1]$.

$$\int_0^1 f(x)dx = \int_0^{1/2} f(x)dx + \int_{1/2}^1 f(x)dx = \frac{1}{2} + \frac{1}{2} = 1$$

Example 2 (Limit of Sum):

Find sum of the series $\frac{n}{n^2} + \frac{n}{n^2+1^2} + \cdots + \frac{n}{n^2+(n-1)^2}$

Solution: General term is $\frac{n}{n^2+r^2}$ where $r = 0, 1, \dots$

$$= \frac{1}{n} \left[\frac{1}{1 + (r/n)^2} \right]$$

Replacing $\frac{1}{n}$ by dx and $\frac{r}{n}$ by x with limits 0 to 1:

$$\text{Sum} = \int_0^1 \frac{dx}{1+x^2} = \tan^{-1}(x) \Big|_0^1 = \frac{\pi}{4}$$

Example 3 (IIT JAM 2023):

Which are Riemann integrable on $[0, 1]$?

- (a) $f(x) = x \sin(1/x)$ if $x \neq 0$, 0 if $x = 0$.
True: Continuous everywhere on $[0, 1]$.
- (b) $f(x) = \int_0^x |\frac{1}{2} - t| dt$.
True: Integrals yield continuous functions.
- (c) $f(x) = x$ if $x \in [0, 1)$, 0 if $x = 1$.
True: Discontinuous at only $x = 1$.
- (d) $f(x) = 1$ if $x \in \mathbb{Q} \cap [0, 1]$, -1 otherwise.
False: Discontinuous at all points. (Dirichlet function).